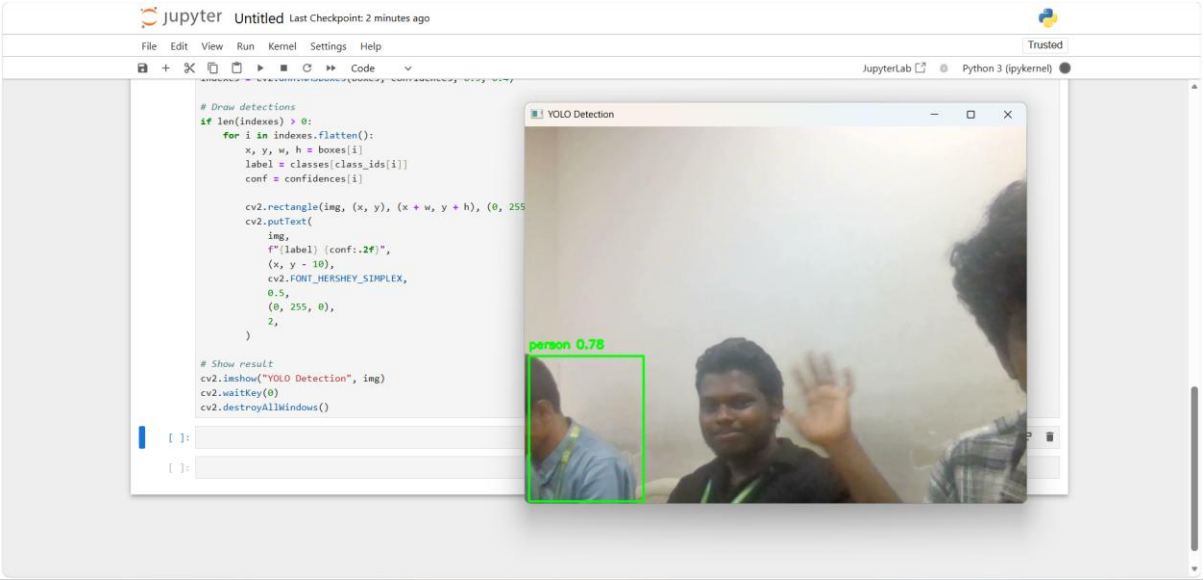
exp 9

```
import cv2
import numpy as np
net = cv2.dnn.readNet("yolov3-tiny.weights", "yolov3-tiny.cfg")
classes = open("coco.names").read().splitlines()
img = cv2.imread("images/person.jpg")

if img is None:
print("Image not found")
exit()

height, width = img.shape[:2]
blob = cv2.dnn.blobFromImage(img, 1/255, (416,416), swapRB=True)
net.setInput(blob)
layer_names = net.getLayerNames()
output_layers = [layer_names[i - 1] for i in net.getUnconnectedOutLayers()]
outputs = net.forward(output_layers)
```

```
boxes = []
confidences = []
class_ids = []

for out in outputs:
for det in out:
scores = det[5:]
class_id = np.argmax(scores)
confidence = scores[class_id]

if confidence > 0.5:
cx, cy, w, h = (det[0:4] * [width, height, width,
height]).astype("int")

x = int(cx - w/2)
y = int(cy - h/2)

boxes.append([x, y, w, h])
```

```
confidences.append(float(confidence))
class_ids.append(class_id)
indexes = cv2.dnn.NMSBoxes(boxes, confidences, 0.5,
0.4)
for i in indexes:

i = i[0] if isinstance(i, (list, tuple, np.ndarray)) else i
x, y, w, h = boxes[i]
label = classes[class_ids[i]]
conf = confidences[i]

cv2.rectangle(img, (x, y), (x+w, y+h), (0,255,0), 2)
cv2.putText(img, f"{label} {conf:.2f}",
(x, y-5), cv2.FONT_HERSHEY_SIMPLEX,
0.5, (0,255,0), 2)
cv2.imshow("YOLO Detection", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```


jupyter Untitled Last Checkpoint: 2 minutes ago
File Edit View Run Kernel Settings Help
Trusted
Code
JupyterLab
Python 3 (ipykernel)
# Draw detections
if len(indexes) > 0:
for i in indexes.flatten():
x, y, w, h = boxes[i]
label = classes[class_ids[i]]
conf = confidences[i]
cv2.rectangle(img, (x, y), (x + w, y + h), (0, 255
cv2.putText(
img,
f"{label} {conf:.2f}",
(x, y - 10),
cv2.FONT_HERSHEY_SIMPLEX,
0.5,
(0, 255, 0),
2,
)
# Show result
cv2.imshow("YOLO Detection", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
[ ]:
[ ]:
YOLO Detection
person 0.78